

Atoh1 Progenitor Analysis

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```
# load packages
library(tidyverse)
library(Seurat)
library(ggplot2)
```

```
setwd("/Users/selinaras/Desktop/")
```

```
# check the working directory
getwd()
```

```
## [1] "/Users/selinaras/Desktop"
```

Load Dataset

```
# load in dataset
atoh1.data <- load("/Users/selinaras/Desktop/Atoh1Progenitors.Rdata")
```

```
# investigate the dataset - look at first 30 cells
atoh1.data[1:30]
```

```
## [1] "SubObj" NA NA NA NA NA NA NA
## [9] NA NA NA NA NA NA NA NA
## [17] NA NA NA NA NA NA NA NA
## [25] NA NA NA NA NA NA
```

```
atoh1 = SubObj
```

```
# what happens if we call the object
head(atoh1, 5)
```

```
##                                orig.ident nCount_RNA nFeature_RNA
## E9.5_GFP.data_AAAGGGCAGGAAAGTG-1    E9.5_GFP      20718      4290
## E9.5_GFP.data_AACAAAGTCATTGTGG-1    E9.5_GFP      31538      5414
## E9.5_GFP.data_AACAACCTCTAGATCG-1    E9.5_GFP      38975      5957
## E9.5_GFP.data_AACAAGAGTTAGCTAC-1    E9.5_GFP      32811      5543
## E9.5_GFP.data_AACCACACACGGCGTT-1    E9.5_GFP      29367      5354
##                                percent.mito nCount_SCT nFeature_SCT color
## E9.5_GFP.data_AAAGGGCAGGAAAGTG-1    0.030504875    19144      4289  GFP
## E9.5_GFP.data_AACAAAGTCATTGTGG-1    0.020102733    19298      5019  GFP
## E9.5_GFP.data_AACAACCTCTAGATCG-1    0.009364978    19423      4864  GFP
## E9.5_GFP.data_AACAAGAGTTAGCTAC-1    0.020877145    19373      5065  GFP
## E9.5_GFP.data_AACCACACACGGCGTT-1    0.031531992    19245      5124  GFP
##                                TimePoint Replicate  Sample percent.blood
## E9.5_GFP.data_AAAGGGCAGGAAAGTG-1    E9.5          1 E9.5 GFP           0
## E9.5_GFP.data_AACAAAGTCATTGTGG-1    E9.5          1 E9.5 GFP           0
```

```
## E9.5_GFP.data_AACAACCTCTAGATCG-1      E9.5      1 E9.5 GFP      0
## E9.5_GFP.data_AACAAGAGTTAGCTAC-1      E9.5      1 E9.5 GFP      0
## E9.5_GFP.data_AACCACACACGGCGTT-1      E9.5      1 E9.5 GFP      0
##                                     seurat_clusters SCT_snn_res.0.8 orig.clusters
## E9.5_GFP.data_AAAGGGCAGGAAAGTG-1      1          0          0
## E9.5_GFP.data_AACAAAGTCATTGTGG-1      1          0          0
## E9.5_GFP.data_AACAACCTCTAGATCG-1      1          0          0
## E9.5_GFP.data_AACAAGAGTTAGCTAC-1      1          0          0
## E9.5_GFP.data_AACCACACACGGCGTT-1      1          0          0
##                                     celltype FullData.category RNA_snn_res.0.5
## E9.5_GFP.data_AAAGGGCAGGAAAGTG-1 Progenitor Progenitor      1
## E9.5_GFP.data_AACAAAGTCATTGTGG-1 Progenitor Progenitor      1
## E9.5_GFP.data_AACAACCTCTAGATCG-1 Progenitor Progenitor      1
## E9.5_GFP.data_AACAAGAGTTAGCTAC-1 Progenitor Progenitor      1
## E9.5_GFP.data_AACCACACACGGCGTT-1 Progenitor Progenitor      1
```

```
# look at the cells in the dataset
```

```
head(colnames(atoh1), 5)
```

```
## [1] "E9.5_GFP.data_AAAGGGCAGGAAAGTG-1" "E9.5_GFP.data_AACAAAGTCATTGTGG-1"
## [3] "E9.5_GFP.data_AACAACCTCTAGATCG-1" "E9.5_GFP.data_AACAAGAGTTAGCTAC-1"
## [5] "E9.5_GFP.data_AACCACACACGGCGTT-1"
```

```
head(Cells(atoh1), 5)
```

```
## [1] "E9.5_GFP.data_AAAGGGCAGGAAAGTG-1" "E9.5_GFP.data_AACAAAGTCATTGTGG-1"
## [3] "E9.5_GFP.data_AACAACCTCTAGATCG-1" "E9.5_GFP.data_AACAAGAGTTAGCTAC-1"
## [5] "E9.5_GFP.data_AACCACACACGGCGTT-1"
```

```
# look at the genes in the dataset
```

```
head(rownames(atoh1), 5)
```

```
## [1] "Xkr4"      "Gm1992"    "Gm37381"   "Rp1"       "Sox17"
```

```
head(Features(atoh1), 5)
```

```
## [1] "Xkr4"      "Gm1992"    "Gm37381"   "Rp1"       "Sox17"
```

```
#what are the slot names
```

```
slotNames(atoh1)
```

```
## [1] "assays"      "meta.data"    "active.assay" "active.ident" "graphs"
## [6] "neighbors"   "reductions"   "images"        "project.name" "misc"
## [11] "version"     "commands"     "tools"
```

This tells us that Features (gene) is the row, and cell (cell info) is in the columns

```
#convert to matrix to get expression data
```

```
atoh1_mat <- as.matrix(atoh1@assays$RNA@data)
```

```
## Warning in asMethod(object): sparse->dense coercion: allocating vector of size
## 1.5 GiB
```

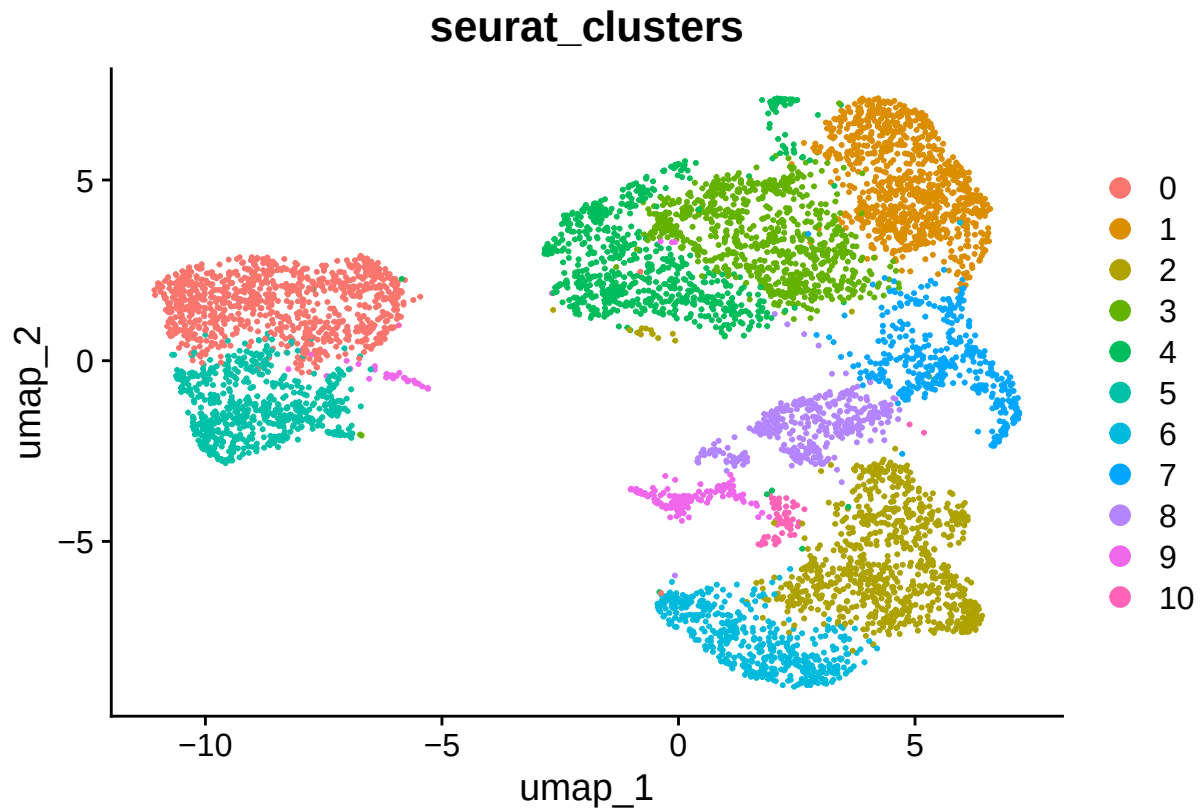
```
atoh1_mat[c("0tx2"), 1:30]
```

```
## E9.5_GFP.data_AAAGGGCAGGAAAGTG-1 E9.5_GFP.data_AACAAAGTCATTGTGG-1
##                                0.0000000                                0.0000000
## E9.5_GFP.data_AACAACCTCTAGATCG-1 E9.5_GFP.data_AACAAGAGTTAGCTAC-1
##                                0.0000000                                0.0000000
```

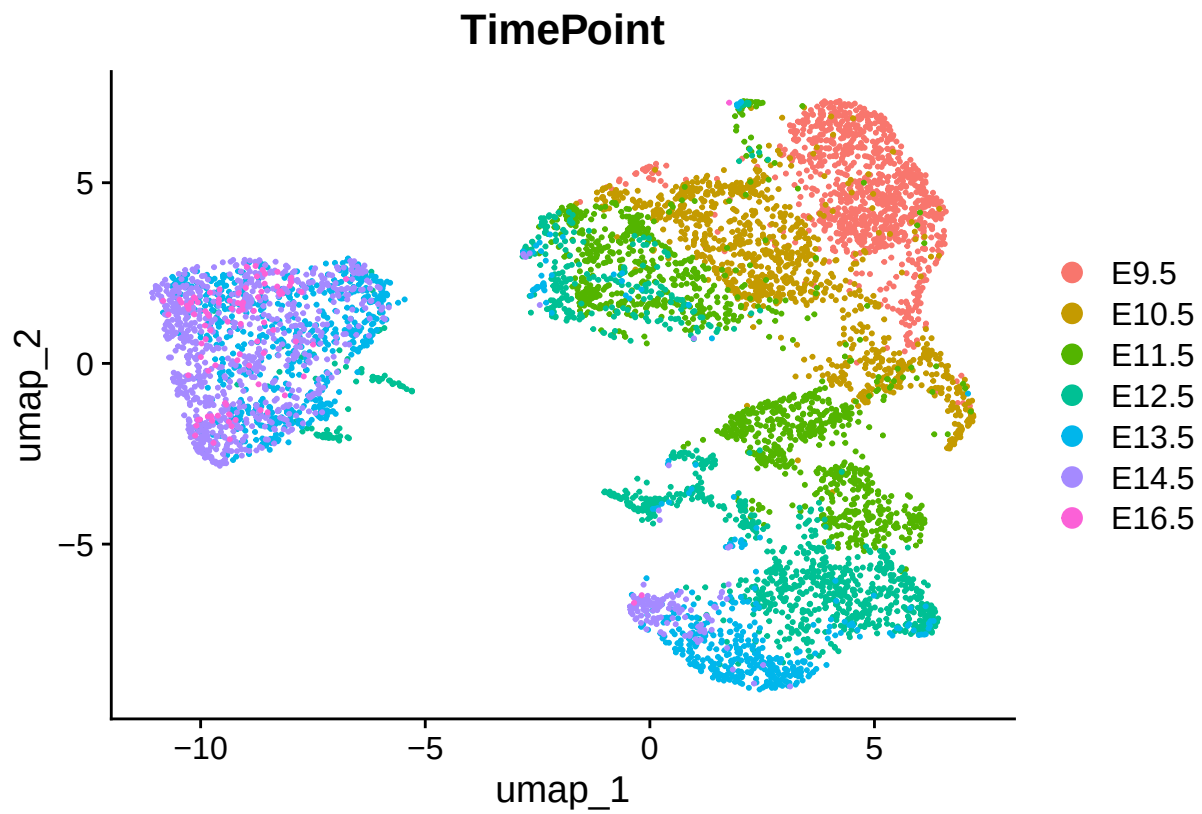
```
## E9.5_GFP.data_AACCACACACGGCGTT-1 E9.5_GFP.data_AACCACATCAAGCCGC-1
## 0.0000000 0.0000000
## E9.5_GFP.data_AACCTTTAGATGCTAA-1 E9.5_GFP.data_AACCTTTAGTGCAACG-1
## 0.0000000 0.0000000
## E9.5_GFP.data_AACCTTTCACGACGAA-1 E9.5_GFP.data_AACGAAAAGTCCCGAC-1
## 0.2689026 0.0000000
## E9.5_GFP.data_AACGTCAAGCCAAGCA-1 E9.5_GFP.data_AACTTCTCATTACCCC-1
## 0.0000000 0.0000000
## E9.5_GFP.data_AACTTCTTCTGAACGT-1 E9.5_GFP.data_AAGAACAAGCATGGGT-1
## 0.0000000 0.0000000
## E9.5_GFP.data_AAGAACAAGCCTTTGA-1 E9.5_GFP.data_AAGAACAGTATATGGA-1
## 0.0000000 0.0000000
## E9.5_GFP.data_AAGAACATCAGAATAG-1 E9.5_GFP.data_AAGACAAGTAAGTTAG-1
## 0.0000000 0.0000000
## E9.5_GFP.data_AAGACAATCCCTCTCC-1 E9.5_GFP.data_AATGGAAGTGACCAA-1
## 0.0000000 0.0000000
## E9.5_GFP.data_AATGGCTTCACATACG-1 E9.5_GFP.data_AATGGCTTCTGAATCG-1
## 0.0000000 0.0000000
## E9.5_GFP.data_ACAAGCTTCGAGAAGC-1 E9.5_GFP.data_ACAAGCTTCTCATAGG-1
## 0.0000000 0.0000000
## E9.5_GFP.data_ACAGCCGGTCCGCAG-1 E9.5_GFP.data_ACATCCCAGTAGTCTC-1
## 0.0000000 0.0000000
## E9.5_GFP.data_ACATCGACAGTGGGTA-1 E9.5_GFP.data_ACATTTTCGTTAGTTCG-1
## 0.0000000 0.0000000
## E9.5_GFP.data_ACCAACATCAACTGAC-1 E9.5_GFP.data_ACCACAACAGTAGTTC-1
## 0.0000000 0.0000000
```

Conducting Data Analysis

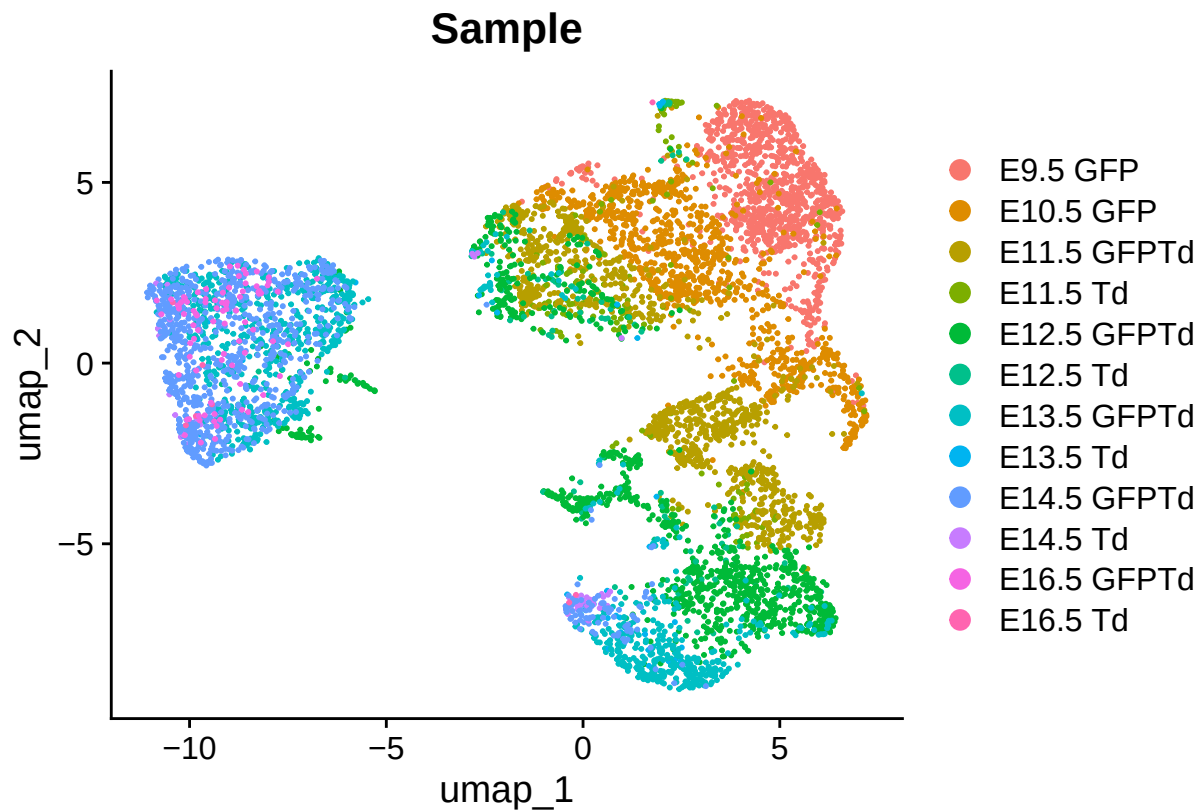
```
# a) by cluster
DimPlot(atoh1, group.by = "seurat_clusters")
```



```
# b) by timepoint  
DimPlot(atoh1, group.by = "TimePoint")
```



```
# c) by sample
DimPlot(atoh1, group.by = "Sample")
```



```
#check if neuronal and glial marker genes are in dataset
markers <- c("Rbfox3", "Map2", "Gfap", "S100b", "Olig1", "Olig2")
markers[markers %in% rownames(atoh1)]
```

```
## [1] "Rbfox3" "Map2" "Gfap" "S100b" "Olig1" "Olig2"
```

```
#assigning vector to neuron and glia markers
neuron_markers <- c("Rbfox3", "Map2")
glia_markers <- c("Gfap", "S100b", "Olig1", "Olig2")
```

```
# count how many cells have an expression value above zero for any neuron or glia marker
neuron_cells <- colSums(atoh1_mat[neuron_markers, ] > 0) > 0
glia_cells <- colSums(atoh1_mat[glia_markers, ] > 0) > 0
```

```
# summarize
summary_counts <- data.frame(
  Category = c("Neuron", "Glial"),
  CellsExpressing = c(sum(neuron_cells), sum(glia_cells))
)
```

```
# view results
summary_counts
```

```
##   Category CellsExpressing
## 1   Neuron           4143
```

```
## 2      Glia      606
```

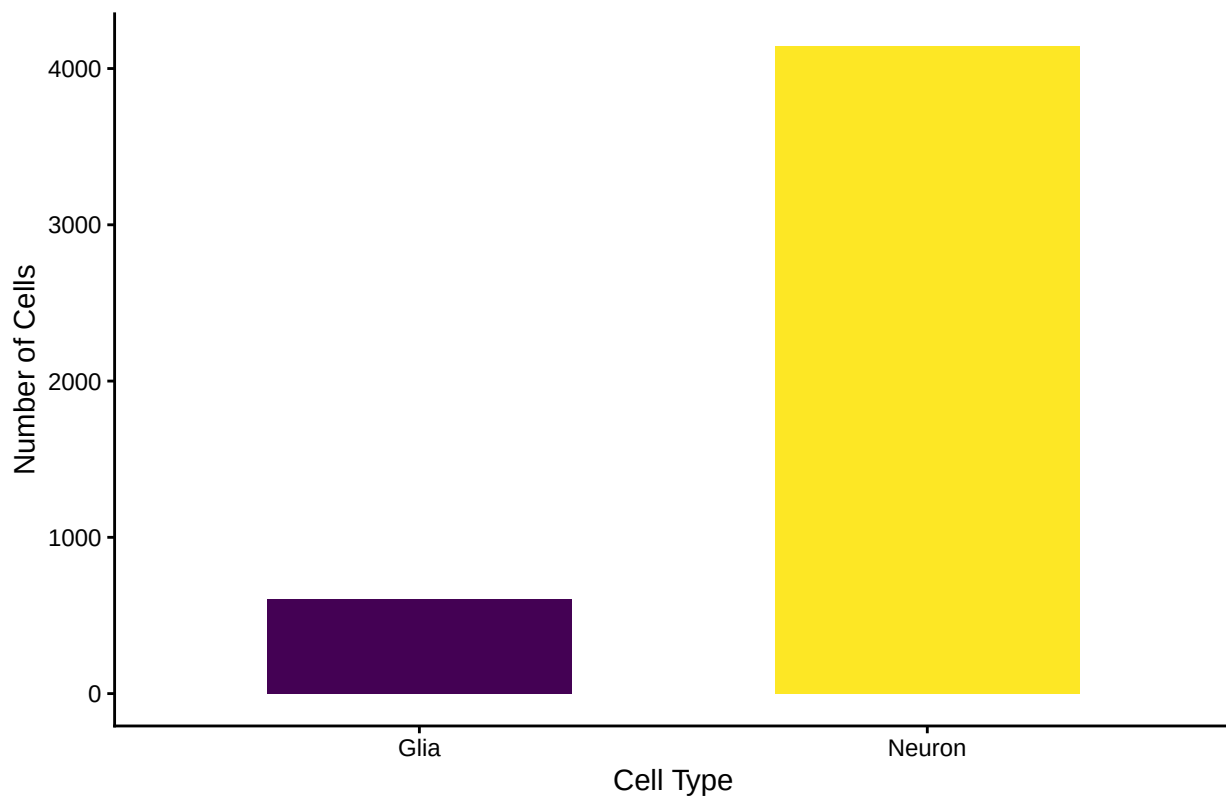
```
# ggplot barplot
```

```
library(viridis)
```

```
## Loading required package: viridisLite
```

```
ggplot(summary_counts, aes(x = Category, y = CellsExpressing, fill = Category)) +  
  geom_col(width = 0.6) +  
  scale_fill_viridis(discrete = TRUE, option = "D") +  
  theme_classic() +  
  labs(  
    title = "Neuron vs Glia Marker Expression",  
    x = "Cell Type",  
    y = "Number of Cells"  
  ) +  
  theme(  
    plot.title = element_text(hjust = 0.5, face = "bold", size = 15),  
    legend.position = "none"  
  )
```

Neuron vs Glia Marker Expression



from the summary counts and graph, we can see that there are more neurons than glia in the dataset

```
# extract Otx2 expression and timepoint info for each cell
```

```
otx2_expr <- otx2_values <- atoh1_mat["Otx2", ]
```

```
otx2_expr <- data.frame(  
  Otx2 = otx2_values,
```

```
  TimePoint = atoh1@meta.data$TimePoint
```

```
)
```

```

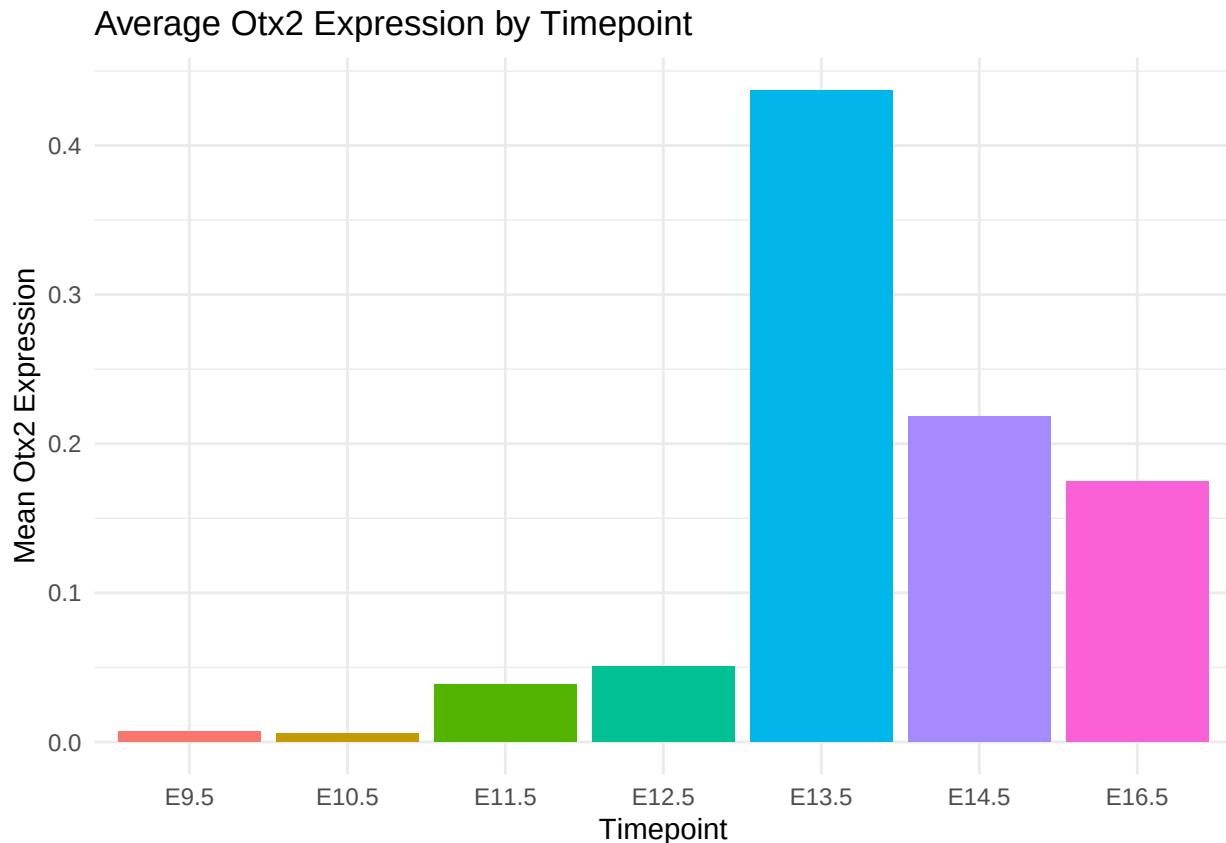
# summarize average expression by timepoint
otx2_summary <- otx2_expr %>%
  group_by(TimePoint) %>%
  summarize(mean_expression = mean(Otx2))

# see which timepoint has the highest average expression
print(otx2_summary)

## # A tibble: 7 x 2
##   TimePoint mean_expression
##   <fct>          <dbl>
## 1 E9.5          0.00695
## 2 E10.5         0.00555
## 3 E11.5         0.0387
## 4 E12.5         0.0507
## 5 E13.5         0.437
## 6 E14.5         0.219
## 7 E16.5         0.175

# plot
ggplot(otx2_summary, aes(x = TimePoint, y = mean_expression, fill = TimePoint)) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  labs(
    title = "Average Otx2 Expression by Timepoint",
    x = "Timepoint",
    y = "Mean Otx2 Expression"
  ) +
  theme(legend.position = "none")

```



the graph shows that the E13.5 timepoint has highest expression of Otx2

Finding cluster biomarkers

```
# set identities to clusters
Idents(atoh1) <- "seurat_clusters"
```

```
# check cluster labels
levels(atoh1)
```

```
## [1] "0" "1" "2" "3" "4" "5" "6" "7" "8" "9" "10"
```

```
# find unique markers for cluster 1
cluster1.markers <- FindMarkers(atoh1, ident.1 = 1)
```

```
## For a (much!) faster implementation of the Wilcoxon Rank Sum Test,
## (default method for FindMarkers) please install the presto package
## -----
## install.packages('devtools')
## devtools::install_github('immunogenomics/presto')
## -----
## After installation of presto, Seurat will automatically use the more
## efficient implementation (no further action necessary).
## This message will be shown once per session
```

```
head(cluster1.markers, n = 5)
```

```
##      p_val avg_log2FC pct.1 pct.2 p_val_adj
## Trim71      0   2.971754 0.930 0.201         0
## Prtg        0   3.438551 0.979 0.254         0
## Gm3764       0  -6.292148 0.104 0.826         0
```

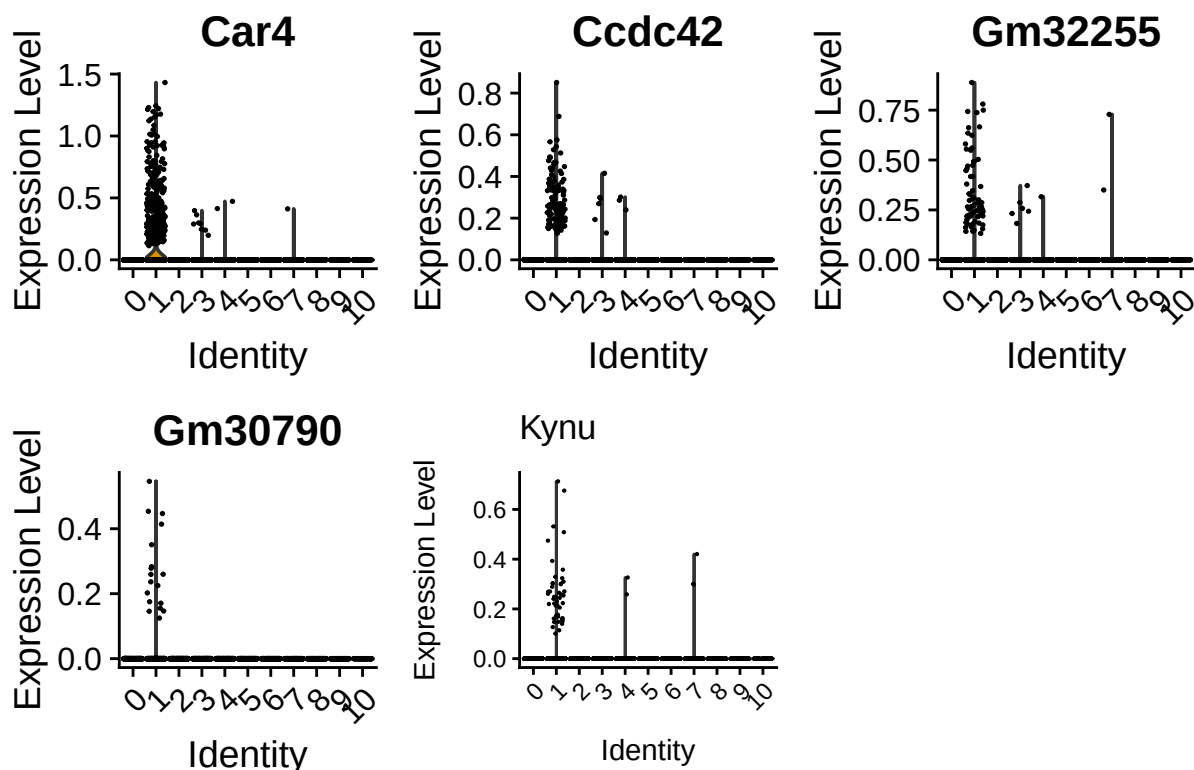


```
## Lin28a      0  4.432306 0.742 0.046      0
## Wfikkn1     0  3.697869 0.701 0.064      0
```

```
# sort by average log2 fold change, since average log fold change measures how much higher the expressi
top_genes <- head(cluster1.markers[order(cluster1.markers$avg_log2FC, decreasing = TRUE), ], 5)
top_genes
```

```
##          p_val avg_log2FC pct.1 pct.2      p_val_adj
## Car4      0.000000e+00  7.723310 0.336 0.002  0.000000e+00
## Ccdc42    2.847169e-142  6.034743 0.129 0.002  8.841598e-138
## Gm32255   3.921740e-81  5.336311 0.078 0.002  1.217857e-76
## Gm30790   8.028254e-25  5.312112 0.019 0.000  2.493094e-20
## Kynu      8.124949e-56  5.037495 0.052 0.001  2.523122e-51
```

```
VlnPlot(
  atoh1,
  features = rownames(top_genes),
  pt.size = 0.1,
  group.by = NULL
) +
  theme_classic() +
  labs(caption = "Top 5 upregulated genes in cluster 1") +
  theme(
    plot.caption = element_text(hjust = 0.5, face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )
```

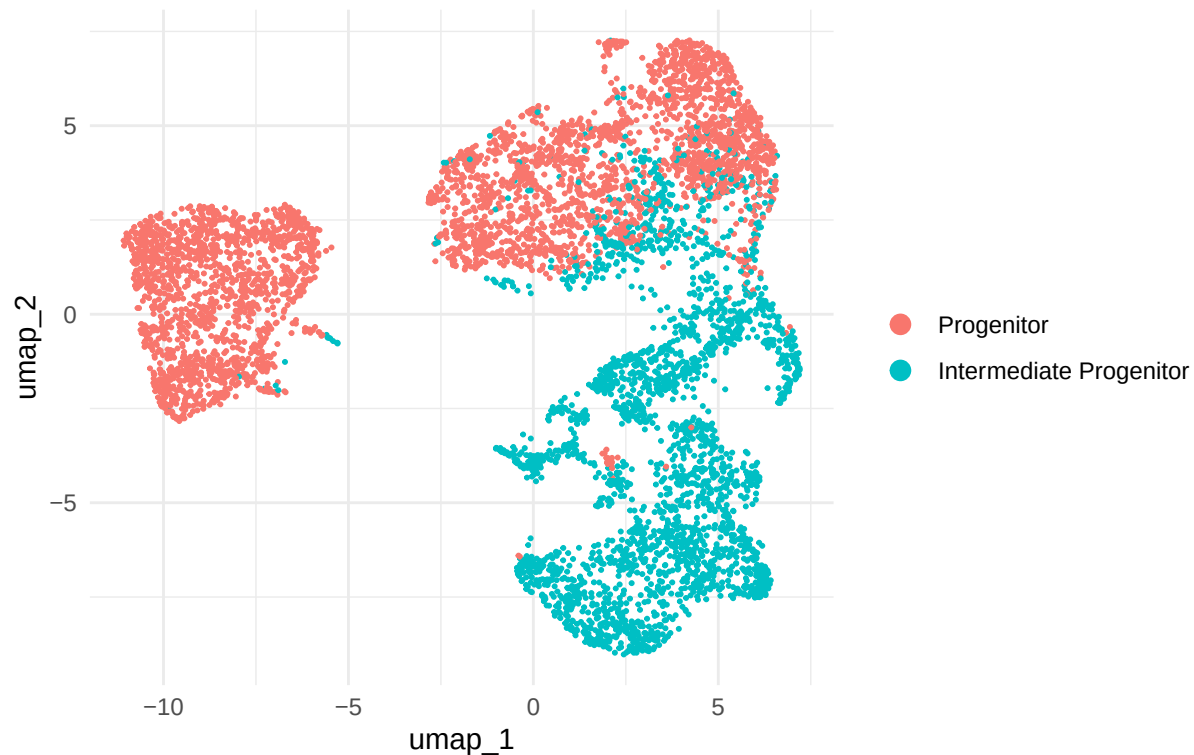


Top 5 upregulated genes in cluster 1

###from the table and graph we can see that Car4 is most upregulated in cluster 1

```
# color by cell type
DimPlot(atoh1, group.by = "celltype") +
ggtitle("Cells Colored by Cell Type") +
theme_minimal() +
theme(
  plot.title = element_text(size = 18, face = "bold"),
  legend.position = "right"
)
```

Cells Colored by Cell Type



```
# set identities to cell type
Idents(atoh1) <- "celltype"
```

```
# check cell type labels
levels(atoh1)
```

```
## [1] "Progenitor" "Intermediate Progenitor"
```

```
# find markers that distinguish cell types
```

```
celltype1.markers <- FindMarkers(atoh1, ident.1 = "Progenitor", ident.2 = "Intermediate Progenitor")
head(celltype1.markers, n = 5)
```

```
##      p_val avg_log2FC pct.1 pct.2 p_val_adj
## Aurkb      0  4.726989 0.918 0.103         0
## Nusap1      0  5.384992 0.935 0.135         0
## Ckap2l      0  4.339770 0.874 0.088         0
## Kif11       0  4.355257 0.889 0.110         0
## Cdk1        0  4.469430 0.957 0.178         0
```

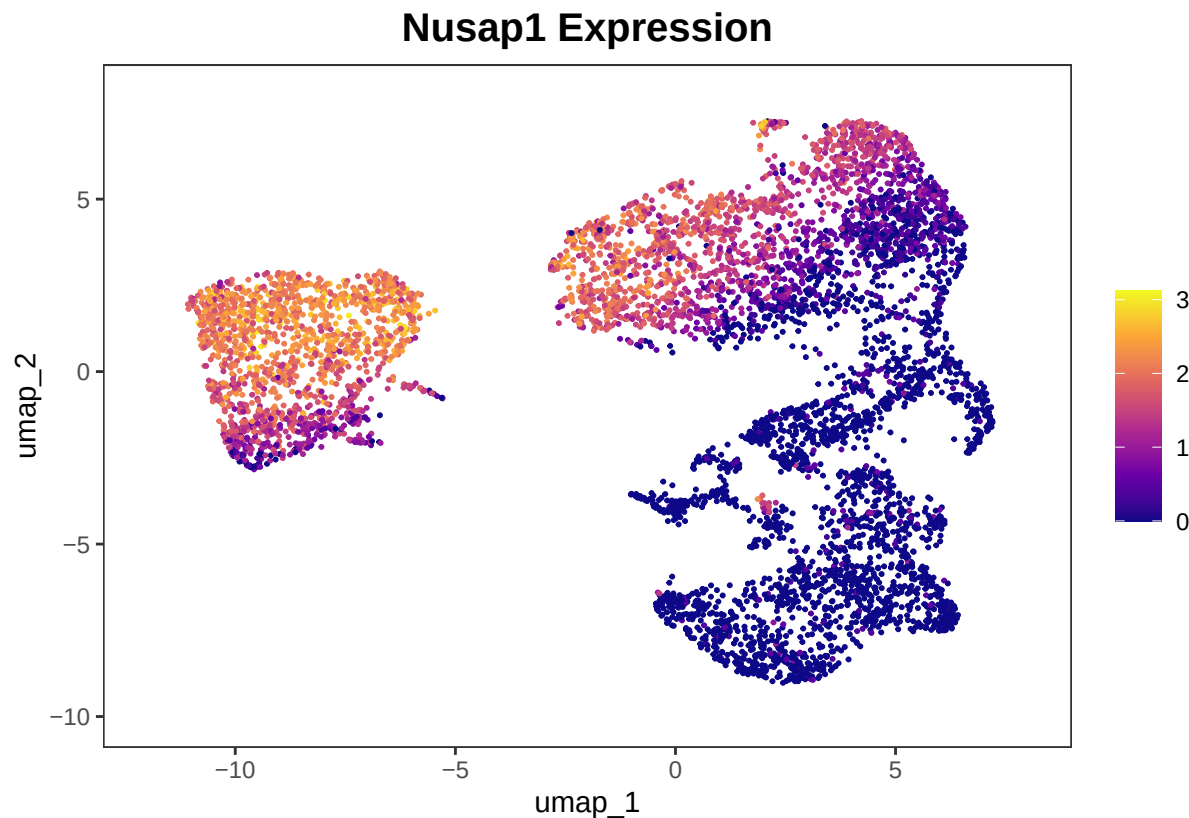
```

#picking Nusap1 and plotting the expression
FeaturePlot(atoh1, features = "Nusap1") +
  scale_color_viridis_c(option = "plasma") +
  ggtitle("Nusap1 Expression") +
  theme_bw() +
  theme(
    panel.grid = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold", size = 15)
  )

```

Scale for colour is already present.

Adding another scale for colour, which will replace the existing scale.



from a average log fold change of 5.38, we can see that Nusap1 is highly expressed in progenitors but lowly expressed in intermediate progenitors

###6 looking at which time points that neuronal (Rbfox3, "Map2") or glial markers ("Gfap", "S100b", "Olig1", "Olig2") show the highest expression

```

# define markers
neuron_markers <- c("Rbfox3", "Map2")
glia_markers <- c("Gfap", "S100b", "Olig1", "Olig2")

# average expression of neuron and glia markers by time point
markers <- c("Rbfox3", "Map2", "Gfap", "S100b", "Olig1", "Olig2")
timepoints <- unique(atoh1$TimePoint)

# empty data frame
df <- data.frame(
  TimePoint = character(),

```

```

Gene = character(),
AvgExpression = numeric()
)

# fill data
for (tp in timepoints) {
  cells <- which(atoh1$TimePoint == tp)

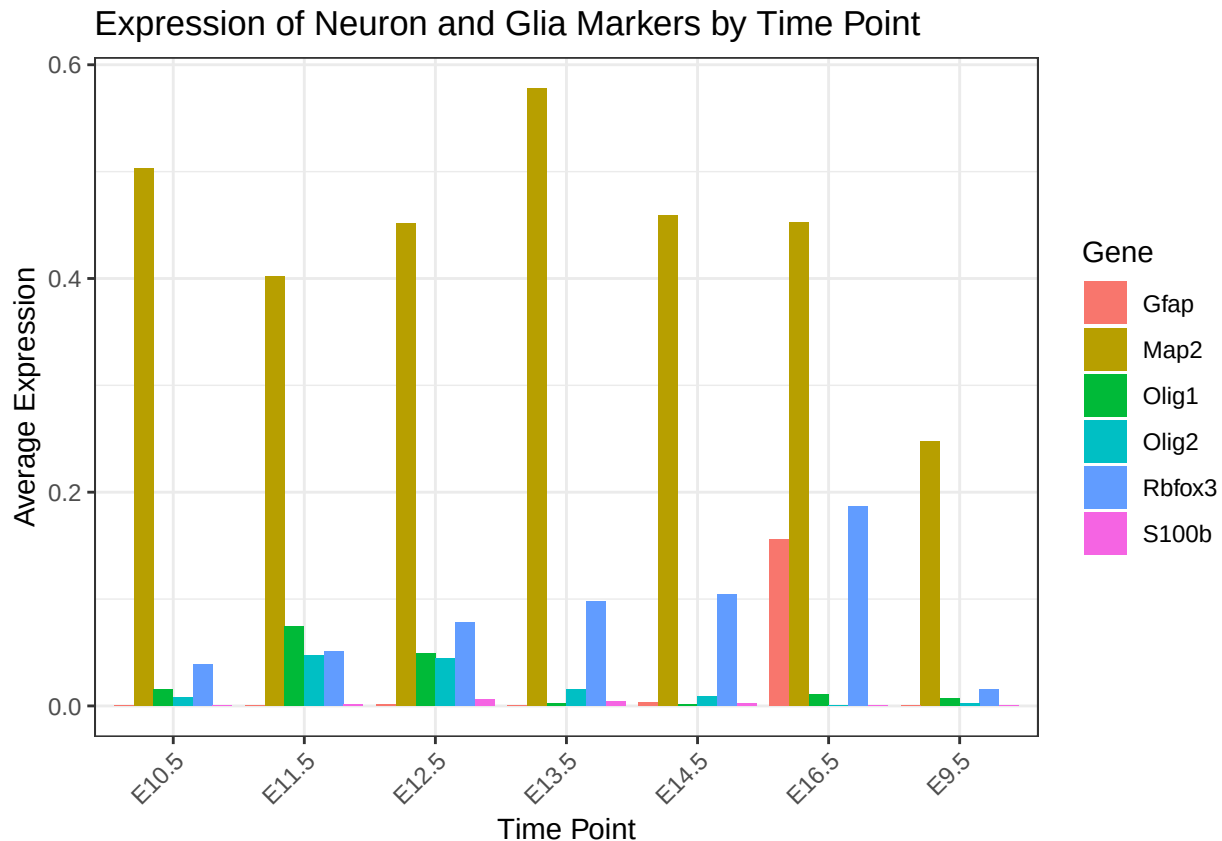
  for (gene in markers) {
    avg_val <- mean(atoh1_mat[gene, cells])

    df[nrow(df) + 1, ] <- list(tp, gene, avg_val)
  }
}

# factor and numeric conversion
df$TimePoint <- factor(df$TimePoint)
df$AvgExpression <- as.numeric(df$AvgExpression)

# plot
ggplot(df, aes(x = TimePoint, y = AvgExpression, fill = Gene)) +
  geom_col(position = "dodge") +
  theme_bw() +
  labs(
    title = "Expression of Neuron and Glia Markers by Time Point",
    x = "Time Point",
    y = "Average Expression"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



The plot shows that Map2 is the dominant marker at all stages, with a peak around E13.5, consistent with widespread neuronal differentiation. Rbfox3 (NeuN) remains low early on but increases steadily and reaches its highest level at E16.5, reflecting late neuronal maturation. Glial markers Olig1 and Olig2 show small signals only at the earliest time points (E10.5–E12.5) and then drop to nearly zero. Gfap, also a glial marker, appears only at E16.5 before returning to zero, briefly overlapping with Rbfox3, which likely reflects a short transitional glial state. S100b remains near zero across all time points, which is expected because S100b is a mature astrocyte marker, and this dataset captures much earlier developmental stages (E9.5–E16.5). The absence of S100b expression indicates that astrocytes have not yet reached maturity during these stages.

now let's look at marker expression over time at progenitors and intermediate progenitors

by progenitor and intermediate progenitor, put them in the same code chunk to compare side-by-side

```
# subset progenitor cells
prog <- subset(atoh1, celltype == "Progenitor")

# define markers
markers <- c("Rbfox3", "Map2", "Gfap", "S100b", "Olig1", "Olig2")

# use only progenitor timepoints
timepoints <- unique(prog$TimePoint)

# empty data frame
df_prog <- data.frame(
```

```

    TimePoint = character(),
    Gene = character(),
    AvgExpression = numeric()
  )

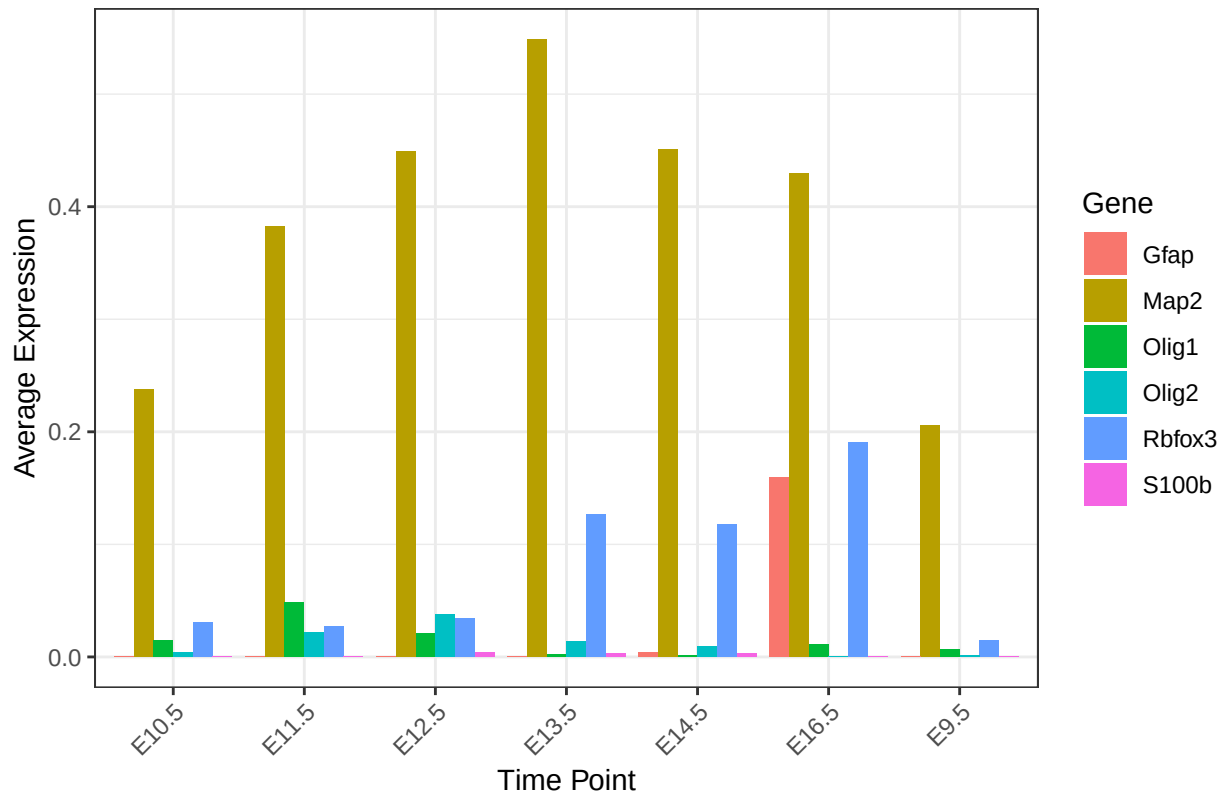
  # fill data
  for (tp in timepoints) {
    cells <- Cells(prog)[prog$TimePoint == tp]
    for (gene in markers) {
      avg_val <- mean(atoh1_mat[gene, cells])
      df_prog[nrow(df_prog) + 1, ] <- list(tp, gene, avg_val)
    }
  }

  # convert types
  df_prog$TimePoint <- factor(df_prog$TimePoint)
  df_prog$AvgExpression <- as.numeric(df_prog$AvgExpression)

  # plot
  ggplot(df_prog, aes(TimePoint, AvgExpression, fill = Gene)) +
    geom_col(position = "dodge") +
    theme_bw() +
    labs(
      title = "Progenitors: Expression of Neuron and Glia Markers by Time Point",
      x = "Time Point",
      y = "Average Expression"
    ) +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

Progenitors: Expression of Neuron and Glia Markers by Time Point



```
### intermediate progenitors
# subset intermediate progenitor cells
int <- subset(atoh1, celltype == "Intermediate Progenitor")

# define markers
markers <- c("Rbfox3", "Map2", "Gfap", "S100b", "Olig1", "Olig2")

# only timepoints present in intermediate progenitors
timepoints <- unique(int$TimePoint)

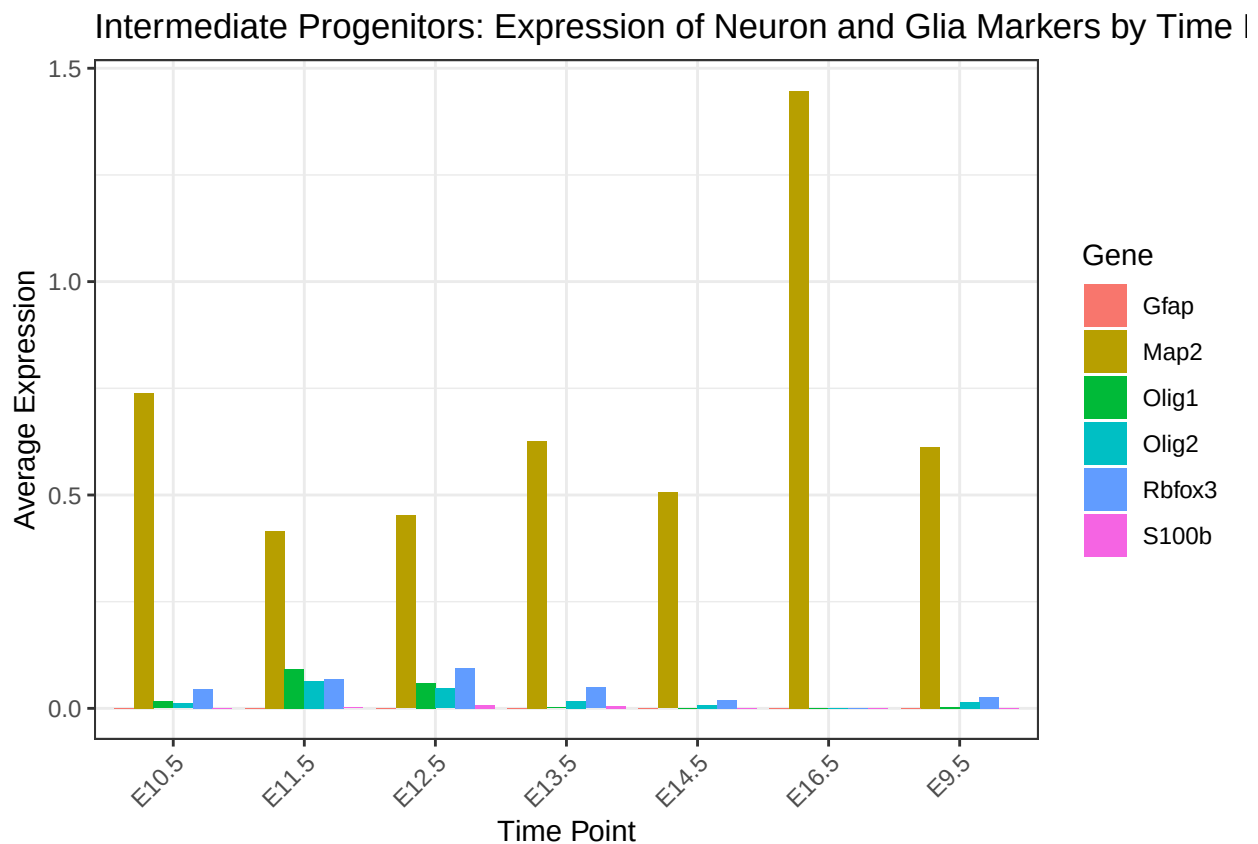
# empty data frame
df_int <- data.frame(
  TimePoint = character(),
  Gene = character(),
  AvgExpression = numeric()
)

# fill data
for (tp in timepoints) {
  cells <- Cells(int)[int$TimePoint == tp]

  for (gene in markers) {
    avg_val <- mean(atoh1_mat[gene, cells])
    df_int[nrow(df_int) + 1, ] <- list(tp, gene, avg_val)
  }
}
```

```
# convert types
df_int$TimePoint <- factor(df_int$TimePoint)
df_int$AvgExpression <- as.numeric(df_int$AvgExpression)

# plot
ggplot(df_int, aes(TimePoint, AvgExpression, fill = Gene)) +
  geom_col(position = "dodge") +
  theme_bw() +
  labs(
    title = "Intermediate Progenitors: Expression of Neuron and Glia Markers by Time Point",
    x = "Time Point",
    y = "Average Expression"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Interestingly, Map2 is slightly higher in intermediate progenitors at E10.5, but it then drops sharply and stays low until a strong peak at E16.5. In progenitors, Map2 rises earlier, peaking around E13.5 before declining. Rbfox3 is essentially absent in intermediate progenitors but increases toward later stages in progenitors, indicating that neuronal maturation occurs mainly within the progenitor group. Olig1 and Olig2 remain low in both populations, but they appear slightly higher in progenitors, and only at the earlier time points, suggesting a brief early window where progenitors retain limited oligodendroglial potential before committing to a neuronal fate. S100b remains near zero across all time points, for the same reasons as mentioned in the first plot from this problem.

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.